Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Innovation project planner

Computer Science and digital literacy give students the tools to design, build and solve problems, not just answer questions about them.

The design thinking cycle

1 Empathise
Understand the people who have the problem.

2 Define
Write the problem in one clear sentence.

3 Ideate
Brainstorm many ideas, then choose the best.

4 Prototype
Build a quick, simple model of your idea.

5 Test
Try it with real users, learn, and improve it.

Computational thinking

Decomposition	Breaking a big problem into smaller parts you can solve.
Pattern recognition	Spotting what parts of a problem have in common.
Abstraction	Ignoring unimportant detail to focus on what matters.
Algorithm	Writing clear step-by-step instructions for the solution.
Iteration	Improving a design by repeating the build-and-test cycle.
User	The person your solution is designed for.

Good digital citizenship

1. Keep passwords private and strong.
2. Think before you post: it can stay online for ever.
3. Respect copyright and credit your sources.
4. Be kind online, as you would be in person.
5. Check facts before you share them.
6. Tell a teacher about anything that worries you online.

Project planner

WHO HAS THE PROBLEM?

THE PROBLEM IN ONE SENTENCE

THREE IDEAS

THE IDEA WE CHOSE, AND WHY

WHAT OUR PROTOTYPE WILL LOOK LIKE

HOW WE WILL TEST IT

WHAT WE LEARNED

WHAT WE WILL IMPROVE

Check yourself

1. Which stage of design thinking comes straight after ‘Ideate’?

2. Breaking a big problem into smaller parts is called…

3. Why do innovators test prototypes with real users?

Answers 1. Prototype: build a quick model of the chosen idea. · 2. Decomposition. · 3. To find out what works and what does not, so they can improve the design.